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ORIGINAL ARTICLE

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Longitudinal study of antinuclear and anticardiolipin antibodies in pregnant women with systemic lupus erythematosus and antiphospholipid syndrome

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Abstract The objective of this study was to perform a longitudinal follow-up of antinuclear antibodies (ANAs) and anticardiolipin antibodies titers (aCL) throughout pregnancy in a group of patients with systemic lupus erythematosus (SLE) and antiphospholipid syndrome (APS) patients during their therapy for the prevention of fetal loss and to examine their relationship with pregnancy outcome. Thirty patients and 15 controls were followed in the study. Fifteen patients had SLE (group I) and 15 had APS (group II, of which seven patients had primary APS and eight had APS secondary to SLE). All patients were receiving therapy for the prevention of fetal loss with prednisone and aspirin as part of an ongoing clinical trial in lupus pregnancy. If there was a history of previous thrombosis, heparin was added. Blood samples were taken at the 1st, 2nd, and 3rd trimesters (T) of pregnancy in order to assess the presence of IgG and IgM aCL antibodies (ELISA), anti-dsDNA (*C. luciliae*) ANAs (HEp-2 cells), and immunospecific antibodies (antiextractable nuclear antigens). We collected 90 samples from patients and 45 samples from healthy controls. Group I (SLE) ANAs were positive in 100% during the 1st T, 67% in the 2nd T, and 67% in the 3rd T, with various immunofluorescence patterns. In five patients, aCL antibodies were detected without a history of APS (one in 1st T, three in 2nd T, and one in 3rd T). Fetal loss was observed in two patients, in one of whom it was associated with nephritis, high titers of ANAs, and anti-dsDNA. Another patient had pulmonary hemorrhage with anti-dsDNA and aCL. In group II, all but one patient with primary APS were negative to ANAs. In secondary APS, by contrast, 6/8 patients (75%) had positive ANAs at least during the 1st T. All seven patients with primary APS and 6/8 with secondary APS had aCL during pregnancy. In 9/15 (60%) patients from the APS group with a history of previous fetal loss, aCL became negative during pregnancy and they had live births. The disappearance of aCL was associated with improved fetal survival (relative risk, or RR, 0.67). ANAs in the control group were positive in 7% at low titers, and all of them were negative for aCL. Despite treatment, ANAs are prevalent during pregnancy in SLE patients and APS secondary to SLE. During pregnancy in SLE, aCL titers may appear. Decreasing titers and/or disappearance of aCL correlated with improved fetal prognosis in a subset of patients with APS.

Keywords Pregnancy in SLE · Antiphospholipid syndrome · Antinuclear antibodies

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Introduction

There is a significantly higher incidence of pregnancy loss in patients with systemic lupus erythematosus (SLE) and antiphospholipid syndrome (APS) than in healthy pregnant women, as well as a high frequency of late miscarriages, intrauterine fetal loss, preterm deliveries, intrauterine growth retardation, antepartum

hemorrhage, and pre-eclampsia [1, 2, 3, 4, 5, 6]. Multiple factors have been associated with these abnormalities, among which are the presence of antiphospholipid antibodies (aPL), lupus activity prior to or during pregnancy, and previous obstetric history of fetal loss. The mechanisms of pregnancy loss in APS are still unknown, and different hypotheses have been proposed as possible causes. Different histologic examinations of the placenta have demonstrated thrombosis, infarction, and even inflammatory changes [7, 8]. Several treatment reports on prevention of pregnancy loss, including corticosteroids, low-dose heparin, heparin, and immunoglobulins, have been used either as single agents or in combination [7, 9, 10, 11]. Also, several studies have also suggested that decreased aPL antibody levels were associated with a successful outcome, and few studies have followed the antinuclear antibodies (ANAs) and aPL antibody levels in pregnant women with these diseases [12, 13]. The present study follows the levels of these antibodies throughout pregnancy in patients who were receiving treatment for the prevention of fetal loss and analyzes the results according to pregnancy outcome.

Patients and methods

Patients

Between March 1998 and March 2000, 58 patients were followed in our Rheumatic Disease Unit and in the Pregnancy Clinic. From this group, 30 patients were referred to the clinic at the time of diagnosis of pregnancy and recruited into the study. Fifteen SLE patients fulfilled the American College of Rheumatism criteria for the classification of the disease [14] and were inactive and consecutively selected (group I). All 15 patients with APS were classified according to the accepted Sapporo's criteria [15]; seven patients had primary APS and eight had secondary APS associated with SLE. All the patients with APS were selected on the basis of history of previous fetal loss (two or more fetal losses). We excluded SLE patients with other well-known causes of recurrent miscarriage or fetal loss, recorded by medical history including anatomical abnormalities, hormonal, infectious, or chromosomally associated disorders.

Controls

We included 15 healthy pregnant women admitted to prenatal control during the same time period. These women had one or more normal term pregnancies with no losses and no history of SLE or other autoimmune conditions. Written voluntary consent to participate in the study was obtained from patients and healthy controls. The hospital-based ethics committee approved the study protocol.

A complete medical history and physical examination were performed when pregnancy was confirmed, and the patients were simultaneously seen by rheumatologists and obstetricians each month until delivery. To follow the fluctuations of ANAs and the levels of anticardiolipin (aCL) antibodies during the course of pregnancy, blood samples were taken at the 1st, 2nd, and 3rd trimesters from all patients. They were frozen and stored at –20°C until all laboratory test procedures were done.

Disease activity of SLE was evaluated according to a physician's global assessment using a scoring system [16] defined as: grade 0 (no activity), grade 1 (mild activity) arthritis, oral ulcers, and malar rash, grade 2 (moderate activity) serositis, fever, and vasculitis, and grade 3 (severe activity) active kidney disease, thrombocytopenia, hemolytic anemia, and neurologic disorder.

Treatment during pregnancy

Patients with SLE without APS received prednisone at the minimal dose required to control SLE plus 85 mg/day of aspirin throughout the pregnancy. All patients with APS (primary and secondary) received prednisone before the 20th week of pregnancy at a dose of 40 mg/day during 4 weeks [11]. The dose was then tapered 10 mg every 4 weeks up to a maintenance dose of 5 mg/day. These patients also received aspirin at a dose of 85 mg/day throughout the pregnancy. If they had a history of previous thrombosis, subcutaneous nonfractionated heparin at a dose of 5,000 IU twice daily was added to the treatment. In both subgroups, drugs were continued during pregnancy until 1 week before labor.

Immunological evaluation

The ANAs were detected by indirect immunofluorescence using HEp-2 cells as substrate (Binding Site, San Diego, Calif., USA) at different dilutions. According to our experience in doing ANAs during the last 20 years, we consider the presence of ANAs when the titer of positive fluorescence is 1:40 or higher. On the other hand, anti-dsDNA antibodies were studied by indirect immunofluorescence on *C. luciliae* (Binding Site), and a positive result was considered when fluorescence of the kinetoplast was present at a dilution of 1:10. Immunospecificity for antibodies to extractable nuclear antigens (ENAs) (Sm, nRNP, SS-B/La, Scl-70, rRNP) was determined by double immunodiffusion. The aCL antibodies were detected by an enzyme-linked immunosorbent assay (ELISA) method using a commercial kit (Corgenix, Westminster, Colo., USA). Results were expressed in IU and, according to the instructions provided by the manufacturer, the cutoff point to consider a positive result was 23 IU for the IgG isotype and 11 IU for the IgM isotype.

Definition of variables

Fetal death was defined as a spontaneous termination of pregnancy after 20 weeks of gestation. Spontaneous abortion was defined as spontaneous termination of pregnancy before 20 weeks of gestation. Pregnancy loss included both spontaneous abortion and fetal death. The term birth was defined as termination of pregnancy with a live birth between the weeks 38 and 40, and premature birth was defined as termination of pregnancy with a live birth between the weeks 21 and 37. Pre-eclampsia was diagnosed if there was an abrupt onset of hypertension and proteinuria after 24 weeks of gestation. Premature rupture of membranes (PROM) was spontaneous rupture of membranes before the onset of labor and delivery. Successful outcome was defined as a pregnancy that ended with a live birth.

Statistical analysis

We used descriptive statistics for continuous variables (mean, SD, and range). To compare aCL antibody levels at different trimesters per patient, we used Wilcoxon's matched pairs test. Proportions were compared using Fisher's exact test. The association between autoantibody levels and pregnancy outcome or disease activity were tested using Spearman's correlation coefficient. P values of 0.05 or less were considered significant.

Results

Demographic and clinical data

Mean age for all patients was 28 ± 4 years (range 22–36), and disease duration was 4 ± 3 years (range 1–16). In group I (SLE), all the patients had skin and joint

involvement, eight had nephritis, serositis was seen in two, another two had pulmonary involvement, and hematologic manifestations were reported in one. In group II (APS primary and secondary), thrombotic episodes were recorded as follows: four patients had one episode, two had two, and one had three. In seven patients, recurrent fetal loss was the only APS clinical manifestation.

Antinuclear and anti-dsDNA antibodies

In Table 1, we show the distribution of ANAs and aCL antibodies in the two groups and controls. In group I, considering all 23 patients with SLE, 17 (74%) had positive ANAs, with a speckled pattern in the majority of cases. Two patients with SLE (9%) had anti-dsDNA antibodies, and four (25%) had positive anti-ENAs. Only one subject from the control group had positive ANAs with negative anti-dsDNA and anti-ENA.

In group II, one patient of primary APS had AAN detected for the first time in the 3rd trimester. In secondary APS, the ANAs were positive in 75% of the cases in the 1st trimester, in 50% in the 2nd trimester, and in 25% in the 3rd trimester.

aCL antibody levels during pregnancy

In 9/15 (60%) patients of the APS group, the aCL antibodies became negative during pregnancy and they had live births (Table 2). The disappearance of aCL was associated with improved fetal survival (RR 0.67). All controls were negative for aCL antibodies. There were no differences in the levels of IgM aCL antibodies among the different trimesters. By contrast, a significant decrease in IgG aCL antibodies levels was observed between the 1st and 3rd trimesters (34.7 ± 19.35 units of anticardiolipin antibodies of IgG class [GPL/U] vs 19.3 ± 16.36 GPL/U, respectively, $P = 0.01$) (Fig. 1). In the SLE group without previous aCL antibodies, these antibodies were detected for the first time in five patients (33%), four with the IgG isotype and one with IgG and IgM isotypes.

Pregnancy outcome

Twenty-six (87%) pregnancies ended successfully. The mean gestational age of the babies was 36 ± 1.8 weeks (range 33–40), and the mean weight at birth was $2{,}337 \pm 670$ g (range 950–3,200). Thirty percent of these babies were premature and 70% were born at term. Among the 30 pregnancies, there were four fetal losses (13%), two in patients with APS (Table 1). From these, one was associated with high and persistent titers of IgG-aCL antibody levels and occurred at the 30th week of gestation. The other fetal loss of this group was related to an accident at the 20th week of pregnancy.

Also, we recorded two pregnancy losses in SLE patients without APS. One of these patients had renal activity at the 12th week of gestation. A remission period was never achieved despite increasing the prednisone dose, and fetal loss occurred at the 24th week of gestation. Another patient developed pulmonary hemorrhage at the 25th week of gestation with high titers of anti-dsDNA and IgG-aCL antibodies. She was treated with pulses of methylprednisolone with good clinical response, but she had a fetal loss 1 week later.

Antibody levels, disease activity, and pregnancy outcome

A statistically significant association between the presence of IgG-aCL antibodies and the risk of fetal loss was documented ($P = 0.04$, RR 7.8, 95% CI 1.1–58). All of these patients had at least one fetal loss. In addition,

Table 1. Distribution of ANAs and aCL antibodies in patients and controls. *APS* antiphospholipid syndrome

Clinical groups	Positive ANAs[a] (%)	Positive aCL (%)	Pregnancy outcome: live births (%) fetal loss (%)
Group I. SLE pregnant women ($n = 15$)			
1st. trimester	15/15 (100)	1/15 (7)	
2nd. trimester	10/15 (67)	3/15 (20)	
3rd. trimester	10/15 (67)	1/15 (7)	13 (87) 2 (13)
Group II. Primary APS pregnant women ($n = 7$)			
1st. trimester	0/7 (0)	7/7 (100)	
2nd. trimester	0/7 (0)	4/7 (57)	
3rd. trimester	1/7 (14)	1/7 (14)	5 (71) 2 (29)[b]
Secondary APS pregnant women ($n = 8$)			
1st. trimester	6/8 (75)	4/8 (50)	
2nd. trimester	4/8 (50)	2/8 (25)	
3rd trimester	2/8 (25)	2/8 (25)	8 (100) 0 (0)
Controls ($n = 15$)			
Healthy women	1/15 (7)	0/15 (0)	15 (100) 0 (0)

[a]The specificity of ANAs in the SLE group was as follows: anti-nDNA 2/15 (13%), anti-nRNP 1/15 (7%), anti-SSB/La 1/15 (7%), anti-rRNP 2/15 (13%)
[b]One of these pregnancy losses was related to an accident

Table 2. Pregnancy outcome in antiphospholipid syndrome (APS)[a] patients whose aCL-IgG became negative. *LB* live births, *ND* not done

Patient	Previous fetal loss	aCL-IgG (GPL/U)[b] positive any trimester	aCL-IgG (GPL/U)[b] negative 3rd trimester	Pregnancy outcome
1	2	24.3	11.0	LB
2	2	23.4	17.0	LB
3	4	34.7	16.2	LB
4	3	25.0	6.26	LB
5	2	38.9	22.5	LB
6	2	ND	20.4	LB
7	2	23.0	10.3	LB
8	2	26.7	10.6	LB
9	3	ND	11.1	LB

[a]APS, primary and secondary
[b]aCL-IgG normal values up to 23 U/GPL

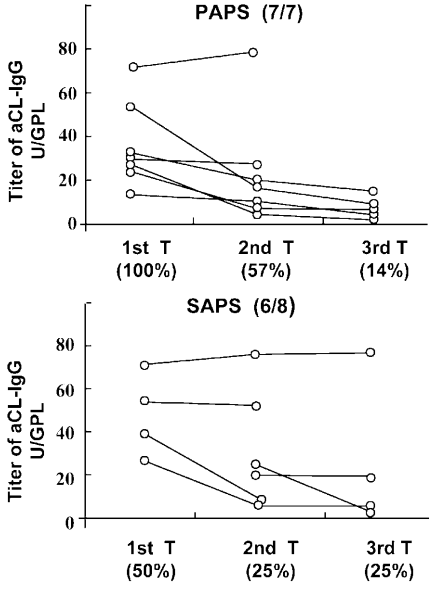


Fig. 1. Course of aCL antibodies in APS patients. The figure shows the titer of aCL-IgG antibodies (U/GPL) during the different trimesters (T) of the pregnancy course in the APS group

there was an association between the presence of anti-dsDNA antibodies and the risk of fetal loss ($P = 0.01$, RR 12, 95% CI 1.7–86). Mild SLE flares did not affect pregnancy outcome ($P > 0.05$); however, a severe disease flare was associated with fetal loss ($P = 0.01$, RR 13.9, 5% CI 1.7–100).

Discussion

We found that in pregnant women with SLE and APS secondary to SLE, ANAs were prevalent despite treatment for disease activity. However, our study suggests that the disappearance of aCL antibodies during treatment to prevent fetal loss may be associated with significant fetal survival in a subset of patients with APS. In addition, we observed that APS might start in some SLE patients during pregnancy. We also support the hypothesis that the presence of ANAs is not associated with disease activity or fetal loss in pregnant women with SLE and APS secondary to SLE. In our control group of healthy pregnant women, aCL antibodies were negative and ANAs were positive in 7% of the cases, similarly to results reported in other studies [17].

An increased level of anti-dsDNA antibodies during pregnancy in SLE patients should alert the physician to the possibility of active lupus nephritis. Elevated levels of anti-dsDNA in these patients seem to correlate also with prematurity, and the association of anti-dsDNA with aCL antibodies during pregnancy may suggest an increased risk of fetal loss [5]. Our results confirmed these clinical observations. We observed fetal loss in two pregnant SLE patients, in one associated with severe nephritis and high titers of anti-dsDNA antibodies, and the other patient developed pulmonary hemorrhage in association with high titers of anti-dsDNA and aCL antibodies without a history of APS. Taking these observations together, we suggest that the presence of anti-dsDNA and aCL antibodies during pregnancy in SLE and APS secondary to SLE are associated with severe complications.

Antiphospholipid antibodies, particularly IgG aCL antibodies and lupus anticoagulant (LA) are associated with fetal loss in SLE and APS and in women with recurrent spontaneous abortion [18, 19, 20]. Therefore, testing these antibodies in SLE and APS patients during pregnancy is very important. In our study we included only the presence of aCL antibodies because LA was not determined in all patients during each trimester of pregnancy.

Different studies have demonstrated an association between high levels of aCL antibodies and adverse fetal outcome in unselected populations of pregnant women [21] and in patients with SLE and APS [22]. However, the significance of low titers or disappearance of these antibodies after treatment is controversial. In this regard, Silver et al. conducted a study on women who had elevated aCL antibody levels and recurrent fetal loss at some point in their history. When stratified by levels of aCL antibodies and followed longitudinally, women in the high-positive aCL antibodies group were more likely to develop one new fetal loss than those in the low-positive aCL antibody groups [23, 24]. Petri et al. established that even low to moderate levels of aCL

antibodies are associated with fetal loss in SLE [25]. In contrast, Silveira et al. found that titers of aCL in patients with primary antiphospholipid syndrome (PAPS) markedly decreased during treatment [13]. However, it is generally agreed that high levels of aCL antibodies are associated with an increased risk of adverse fetal outcome in SLE. If untreated, over 80% of women with a history of prior fetal death and high levels of IgG aCL antibodies will experience a fetal death in their next pregnancy [26].

Therapy for the prevention of fetal loss in both SLE and primary APS remains controversial [10, 20, 27]. Although definitive clinical trials have not been done [28, 29], we used generally accepted protocols during the study period. We included patients with SLE, primary APS, and APS secondary to SLE. The steroid treatment protocol was with 40 mg of prednisone given for only 4 weeks, and the dose was then tapered 10 mg every 4 weeks up to a maintenance dose of 5 mg/day. We observed a high live birth rate (87%) in patients with SLE and with primary or secondary APS. In group II, patients had histories of at least one fetal loss. All patients with APS secondary to SLE had live births after treatment. Interestingly, in 6/7 patients with primary APS, aCL antibodies became negative, five of whom had live births, and only one patient had a fetal loss associated with persistent high titers of aCL antibodies. In a limited experience in two patients with SLE and previous fetal loss associated with high titers of aCL antibodies, Erlendsson et al. observed in one subsequent pregnancy treated with prednisolone a disappearance of aCL antibody titers and normal pregnancy outcome [8]. Our study suggests that prednisone therapy may produce decreased levels and disappearance of aCL antibodies associated with live births in a selected group of patients with recurrent fetal loss.

Although the best treatment during pregnancy in SLE or APS is uncertain at this point, most physicians prefer treatment with heparin or heparin (fractionated) and low-dose aspirin [28, 30]. A recent guideline for the treatment of pregnant patients with aPL has been proposed [20]. On the other hand, it is well known that the combination of prednisone and heparin increases the risk of osteoporosis and osteonecrosis [31, 32, 33]. Following our treatment protocol, we have not observed severe complications in pregnant patients treated with prednisone, heparin, and aspirin. The high steroid dosage used for a short time was well tolerated by all patients. The reason could be related to a relatively rapid decrease in the prednisone dose, needed to maintain fetal survival and to decrease or even suppress aCL levels [34].

Finally, in group I (SLE patients), we observed five patients with positive aCL antibodies without previous histories of APS (one of them with fetal loss). This finding suggests that aCL may appear during pregnancy. Supporting this idea, there are several studies on the pathophysiology of antiphospholipid antibodies and their relationship with pregnancy loss. Also, antiphospholipid antibodies have been implicated in the fusion of cytotrophoblast to syncitial cells, a critical step in the development and maturation of the villous trophoblast [35, 36, 37].

In summary, ANAs were prevalent during pregnancy in patients with SLE and APS secondary to SLE, despite treatment. After treatment to prevent fetal loss, the disappearance of aCL antibodies could be associated with improved fetal prognosis in a subset of patients with APS. Additional studies are necessary to confirm these clinical observations.

References

1. Urowitz MB, Gladman DD, Farewell VT, McDonald J (1993) Lupus and pregnancy studies. Arthritis Rheum 36:1392–1397
2. Lockshin MD, Qamar T, Druzin ML (1987) Hazards of lupus pregnancy. J Rheumatol 14:214–217
3. Birdsall M, Pattison N, Chamley L (1992) Antiphospholipid antibodies in pregnancy. Aust N Z J Obstet Gynaecol 32: 328–330
4. Branch DW, Silver RM (1996) Criteria for antiphospholipid syndrome: early pregnancy loss, fetal loss, or recurrent pregnancy loss? Lupus 5:409–413
5. Khamashta MA, Ruíz-Irastorza G, Hughes GRV (1997) Systemic lupus erythematosus flares during pregnancy. Rheum Dis Clin N Am 1:15–29
6. Lynch A, Silver R, Emlen W (1997) Antiphospholipid antibodies in healthy pregnant women. Rheum Dis Clin N Am 23: 55–70
7. Rosove MH, Tabsh K, Wasserstrum N, Howard P, Hanh BH, Kalunian KC (1990) Heparin therapy for pregnant women with lupus anticoagulant or anticardiolipin antibodies. Obstet Gynecol 75:630–634
8. Erlendsson K, Steinsson K, Jöhannsson JH, Geirsson RT (1993) Relation of antiphospholipid antibody and placental bed inflammation vascular changes to the outcome of pregnancy in successive pregnancies of 2 women with systemic lupus erythematosus. J Rheumatol 20:1779–1785
9. Rai R, Cohen H, Dave M, Regan L (1997) Randomised controlled trial of aspirin plus heparin in pregnant women with recurrent miscarriage associated with phospholipid antibodies (or antiphospholipid antibodies). BMJ 311:253–257
10. Cowchock SF, Reece AE, Balaban D, Branch DW, Plouffe L (1992) Repeated fetal losses associated with antiphospholipid antibodies: a collaborative randomized trial comparing prednisone with low-dose heparin treatment. Am J Obstet Gynecol 166:1318–1323
11. Silveira LH, Hubble CL, Jara LJ, Saway S, Martínez-Osuna P, Seleznick MT (1992) Prevention of anticardiolipin antibody-related pregnancy losses with prednisone and aspirin. Am J Med 93:403–411
12. Tomer Y, Viegas OAC, Swissa M, Koh SCL, Shoenfeld Y (1996) Levels of lupus autoantibodies in pregnant SLE patients: correlations with disease activity and pregnancy outcome. Clin Exp Rheumatol 14:275–280
13. Silveira LH, Jara LJ, Espinoza LR (1996) Transient disappearance of serum antiphospholipid antibodies can also be due to prednisone therapy. Clin Exp Rheumatol 14:217–226
14. Tan EM, Cohen AS, Fries JF, Masi AT, McShane DJ, Rothfield NF et al (1982) The 1982 revised criteria for the classification of systemic lupus erythematosus. Arthritis Rheum 25:1271–1277
15. Wilson WA, Gharavi AE, Koike T, Lockshin MD, Branch WE, Piette J-C, Brey R, Derksen R, Harris N, Hughes GRV, Tripplet DA, Khamashta MA (1999). International consensus

statement on preliminary classification criteria for definite antiphospholipid syndrome. Arthritis Rheum 42:1309–1311
16. Jara Quezada L, Graef A, Lavalle C (1991) Prolactin and gonadal hormones during pregnancy in systemic lupus erythematosus. J Rheumatol 18:349–353
17. García De La Torre I, Hernández-Vázquez L, Angulo-Vazquez J, Romero-Ornelas A (1984) Prevalence of antinuclear antibodies in patients with habitual abortion and in normal and toxemic pregnancies. Rheumatol Int 4:87–89
18. Parke AL, Wilson D, Maier D (1991) The prevalence of antiphospholipid antibodies in women with recurrent spontaneous abortion, women with successful pregnancies, and women who have never been pregnant. Arthritis Rheum 34:1231–1235
19. Lockshin MD, Druzin ML, Goei S (1985) Antibody to cardiolipin as a predictor of fetal distress or death in pregnant patients with systemic lupus erythematosus. N Engl J Med 313:152–156
20. Derksen RHWM, De Groot PG, Nieuwenhuis HK, Christiaens GCML (2001) How to treat women with antiphospholipid antibodies in pregnancy? Ann Rheum Dis 60:1–3
21. Perez MC (1991) Anticardiolipin antibodies in unselected pregnant women. Relationship to fetal outcome. J Perinatol 11:33–36
22. Harris EN, Cahn JKH, Asherson RA, Aber VR, Gharavi AE, Hughes GRV (1986) Thrombosis, recurrent fetal loss and thrombocytopenia. Predictive value of the anticardiolipin antibody test. Arch Intern Med 146:2153–2156
23. Silver RM, Draper ML, Scott JR (1994) Clinical consequences of antiphospholipid antibodies: an historic cohort study. Obstet Gynecol 83:372–378
24. Silver RM, Porter TF, van Leeuwen Y, Jeng J, Scott JR, Branch DW (1996) Anticardiolipin antibodies: clinical consequences of ''low titers.'' Obstet Gynecol 87:494–500
25. Petri M, Howard D, Repke J, Goldman DW (1992) The Hopkins Lupus Pregnancy Center: 1987–1991 update. Am J Reprod Immunol 28:188
26. Welch S, Branch W (1997) Antiphospholipid syndrome in pregnancy. Obstetric concerns and treatment. Rheum Dis Clin N Am 23:71–84
27. Espinoza LR (1996) Antiphospholipid antibody syndrome: treatment. Lupus 5:456–457
28. Petri M (2000) Treatment of the antiphospholipid syndrome: progress in the last five years? Curr Rheumatol Rep 2:256–261
29. Laskin CA, Bombardier C, Hannah ME, Mandel FP, Knox RJW, Farewell V (1997) Prednisone and aspirin in women with autoantibodies and unexplained recurrent fetal loss. N Engl J Med 337:148–153
30. Cowchock SF (1996) Prevention of fetal death in the antiphospholipid antibody syndrome. Lupus 5:467–472
31. Dykman TR, Gluck OS, Murphy WA, Hahn TJ, Hahn BH (1985) Evaluation of factors associated with glucocorticoid-induced osteopenia in patients with rheumatic diseases. Arthritis Rheum 28:361–368
32. Zizic TM, Marcoux C, Hungerford DS, Stevens MB (1985) Corticosteroid therapy associated with ischemic necrosis of bone in systemic lupus erythematosus. Am J Med 79:596–604
33. Nelson-Piercy C (1997) Heparin-induced osteoporosis in pregnancy. Lupus 6:500–504
34. Ruíz-Irastorza G, Kamashta M, Hughes GRV (1998) Treatment of the antiphospholipid syndrome in pregnancy. Scand J Rheumatol Suppl 107:44–47
35. Tripplet DA (1993) Antiphospholipid antibodies and thrombosis: a consequence, coincidence or cause? Arch Pathol Lab Med 117:78–88
36. Katsuragawa H (1995) Monoclonal antiphosphatidylserine antibody reactivity against human first-trimester placental trophoblast. Am J Obstet Gynecol 172:1592–1596
37. Rand JH (1997) Pregnancy loss in the antiphospholipid-antibody syndrome: a possible thrombogenic mechanism. N Engl J Med 337:154–160